

MD SHIFATUL AHSAN APURBA

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OBJECTIVE

AI and data science researcher and Ph.D. candidate in Biomedical Health Informatics and Data Science at the University of Alabama at Birmingham, with research interests in generative modeling, computer vision, representation learning, and large language models. Research focuses on developing AI-driven methods for image reconstruction, multimodal learning, and evidence-based analysis. Seeking research and applied AI/ML opportunities to contribute to reliable, efficient, and impactful intelligent systems across biomedical, scientific, and broader real-world applications.

EDUCATION

Ph.D. Candidate, Biomedical Health Informatics Department of Biomedical Informatics and Data Science University of Alabama at Birmingham CGPA: 4.00 out of 4.00 (Current)	Expected 2028
Master of Computer Science and Engineering , BRAC University CGPA: 3.73 out of 4.00	2025
Bachelor of Computer Science and Engineering , BRAC University CGPA: 3.74 out of 4.00	2022

SKILLS

Technical Skills	Python, Java, ASP.NET, SQL, Machine Learning, Power BI, Power Automate, Power Apps
Advanced AI and Data Analysis	TensorFlow, PyTorch, Scikit-learn, OpenCV, Neural Networks, Transformers, Generative Models (Diffusion Models, GANs, MeanFlow Models), Reinforcement Learning, Large Language Model Fine-Tuning, Prompt Tuning and Instruction Tuning of LLMs, Vision-Language Models (OpenAI CLIP, SigLIP), Medical Image Processing, Radiomics, Pandas, NumPy, Matplotlib, Data Pre-processing, Model Training, Evaluation, and Optimization
Soft Skills	Leadership, Communication, Teamwork, Problem-Solving, Adaptability, Mentorship
Others	Adobe Photoshop, Adobe Illustrator, Adobe Premier Pro

EXPERIENCE

Graduate Research Assistant (<i>Part-time</i>) Dept. of Biomedical Informatics and Data Science, University of Alabama at Birmingham	Aug 2024 - Present <i>Birmingham, AL</i>
- Conduct research on generative AI methods for CT image standardization, reconstruction, and medical image analysis. - Build and evaluate deep learning pipelines using clinical imaging datasets, focusing on image quality, structural similarity, and radiomic feature consistency. - Support literature review, experimental design, data analysis, and manuscript preparation for biomedical imaging research projects.	
Specialist, IT Core Applications (<i>Full-time</i>) Robi Axiata PLC	Feb 2024 - Jul 2024 <i>Dhaka, Bangladesh</i>
- Upgraded capacity by 20% for CBS voucher card project implementing advanced SUSE Linux system integrating with APIs and Gauss DB in the system. - Revamped CDR (call details record) mediation system while saving 30% storage.	

- Automated monthly capacity planing and reporting for 3 major charging systems.

Debate Coach (*Part-time*)

Maple Leaf International School

Oct 2022 - Jul 2024

Dhaka, Bangladesh

- Teaching debate to 150+ students (Grade 6-12) involves introducing structure, research skills, persuasive and logical reasoning, public speaking, effective rebuttals, time management, critical listening, respectful dialogue, practice debates, debate etiquette, ethics, and evaluation.

Associate Software Engineer (*Full-time*)

British American Tobacco Bangladesh

Jan 2023 - Jan 2024

Dhaka, Bangladesh

- Launched 21 projects using Power Platform, ASP.NET, PHP Frameworks and ReactJS.
- Saved 11,000+ man-hours and significant operating costs per year including environmental sustainability with automated and paperless initiatives.
- Initiated citizen development program and impacted 700+ employees.

Intern, Digital Business Solutions (*Internship*)

British American Tobacco Bangladesh

Sep 2022 - Dec 2022

Dhaka, Bangladesh

- Coordinated IT projects collaborating with 3 different end-market.
- Arranged 10+ IT workshops for internal employees.
- Built company's guesthouse reservation and management app.

Undergraduate Teaching Assistant (*Part-time*)

School of Data and Science, BRAC University

Nov 2021 - Aug 2022

Dhaka, Bangladesh

- Taught Matlab, Mathematica, and Python and graded scripts of 250+ Students.

Intern, Core Network Service Management (*Internship*)

Banglalink Digital Communication Ltd.

Oct 2021 - Dec 2021

Dhaka, Bangladesh

- Planned data-driven network routing for three nodes spanning two divisions within the country.

RESEARCH AND PUBLICATION

Research Project. RA-CMF: Region-Adaptive Conditional MeanFlow for CT Image Reconstruction (Accepted at ICIBM 2026 Conference) ([Read here](#))

Research Project. A Hybrid DenseNet-Vision Transformer Framework for Diabetic Retinopathy Classification (Accepted at ICCSDFAI 2026 Conference) ([Read here](#))

Research Project. A model to guide successful living with venous thromboembolism (VTE): Recommendations derived from the qualitative phase of a study of diagnostic Delay of VTE (DOVE) ([Read here](#))

Research Project. Unveiling Shadows: A Data-driven Insight on Depression Among Bangladeshi University Students ([Read here](#))

Research Project. Development of an OCR-Based System for Reading Text in Dull Conditions (Presented at IEEE R10 HTC 2023 Conference)

Survey Paper. A Comprehensive Survey on Optimizing Storage Models, Data Layouts, and System Catalogs ([Read here](#))

PROJECT

Personal Project. An Interactive Research Workspace for Provenance-Preserving Biomedical Evidence ([Visit here](#))
Developed an LLM-powered interactive research workspace for provenance-preserving biomedical evidence exploration. The system integrates multiple biomedical APIs, including RxNorm, DailyMed, openFDA, ClinicalTrials.gov, ChEMBL, Open Targets, and PubChem, with uploaded scientific literature to support natural-language evidence retrieval, cross-source reasoning, and structured synthesis. It separates LLM-based interpretation from deterministic evidence composition to improve reproducibility and maintain traceable source provenance.

EXTRA-CURRICULAR ACTIVITIES

Debating. Actively participated in and judged national and international debate tournaments, breaking at major events such as ABP 2021 and ADTI Indo Pacific Pre-WUDC 2021, achieving 2x quarter-finalist positions at Uhuru Worlds, and securing over 15 championship wins.

Filmmaking. Involved in filmmaking with achievements including the nomination of Bakshobondi as the 2nd Best Film at International Children's Film Festival 2017 and as the Best Edited Film at Sylhet International Film Fest 2017, along with the official selection of Oddhayer Sesh at International Children's Film Festival 2015.

Hackathon. Participated in several hackathons, including Robi Datathon 2.0, where placed in the top 20%, and developed a Covid Update and Alerts application during the pandemic, earning the 2nd runners-up award at MIST National ICT Fest 2021.

LEADERSHIP

- Led the BRAC University Debate Club as President, facilitating over 500 students and providing necessary resources, which helped the club secure the World University Debating Championship 2022 title. Convened 1 international and 3 national tournaments during my tenure.
- Led student management for residential semester at BRAC university, initiating an automated meal management system that streamlined operations. Coordinated cultural events such as RS Day and Cultural Night to enhance student engagement.
- Served as a mentor for the first-year advising team at BRAC University, providing guidance to over 50 students on course selection, academic planning, and advising support. Fostered a supportive environment to help students navigate their academic journey and make informed decisions for their future.